

H/W #1

u 22 10 251

page 814 ex 13

$$(4, 8, -1) (3, 8, 1)$$

$$r = \sqrt{(3-4)^2 + (8-8)^2 + (1+1)^2} = \sqrt{30}$$

$$(x-3)^2 + (y-8)^2 + (z-1)^2 = 30$$

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$$|u| = |v| = 1 \quad u + v + w = \vec{0} \quad |w| = ?$$

$$w = -u - v$$

$$w = (-u) + (-v)$$

$$(-u)^2 + (-v)^2 = |w|^2 \quad |-u| = |u| = 1$$

$$|w| = \sqrt{1 + 1} = \sqrt{2} \quad |-v| = |v| = 1$$

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$$a = i + 2j - 2k$$

$$b = 4i - 3k$$

$$a = \sqrt{1^2 + 2^2 + (-2)^2} = \sqrt{9} = 3$$

$$b = \sqrt{4^2 + 0^2 + (-3)^2} = \sqrt{25} = 5$$

$$a \cdot b = 1 \cdot 4 + 2 \cdot 0 + (-2) \cdot (-3) = 10$$

$$\cos \theta = \frac{a \cdot b}{a \cdot b} = \frac{10}{3 \cdot 5} = \frac{2}{3} \quad \theta \approx 48^\circ$$

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$$u = i + 5j - 2k$$

$$v = 3i - 8j$$

$$w = 5i + 8j - 4k$$

$$u \cdot (v \times w) = \begin{vmatrix} 1 & 5 & -2 \\ 3 & -8 & 0 \\ 5 & 8 & -4 \end{vmatrix} = 1 \begin{vmatrix} -10 & 30 \\ 8 & -4 \end{vmatrix} - 5 \begin{vmatrix} 30 & 30 \\ 5 & -4 \end{vmatrix} +$$

$$+ (-2) \begin{vmatrix} 3 & -8 \\ 5 & 8 \end{vmatrix} = 4 + 60 - 64 = 0 //$$

coplanar

page 848 ex N 13

~~ex 13~~ $(-4, -6, 1)$ and $(-2, 0, -3) //$

$$v_1 = (-2 - (-4), 0 - (-6), -3 - 1) = (2, 6, -4)$$

$$v_2 = (5 - 10, 3 - 18, 14 - 4) = (-5, -15, 10)$$

$$-5(1, 3, -2)$$

$$v_1 = 2(1, 3, -2)$$

$$v_2 = -5(1, 3, -2)$$

$$v_2 = -\frac{5}{2} v_1 \text{ parallel.}$$